

Saint Petersburg State
University Graduate School of Management

End-to-end Sample ML Project course

Report 3: Model Training

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Model 1: Linear Regression

Linear Regression - a baseline model that assumes a linear relationship between features and the target variable. Minimizes the sum of squared residuals.

Grid Search Parameters

Parameter	Values Tested
fit_intercept	[True, False]

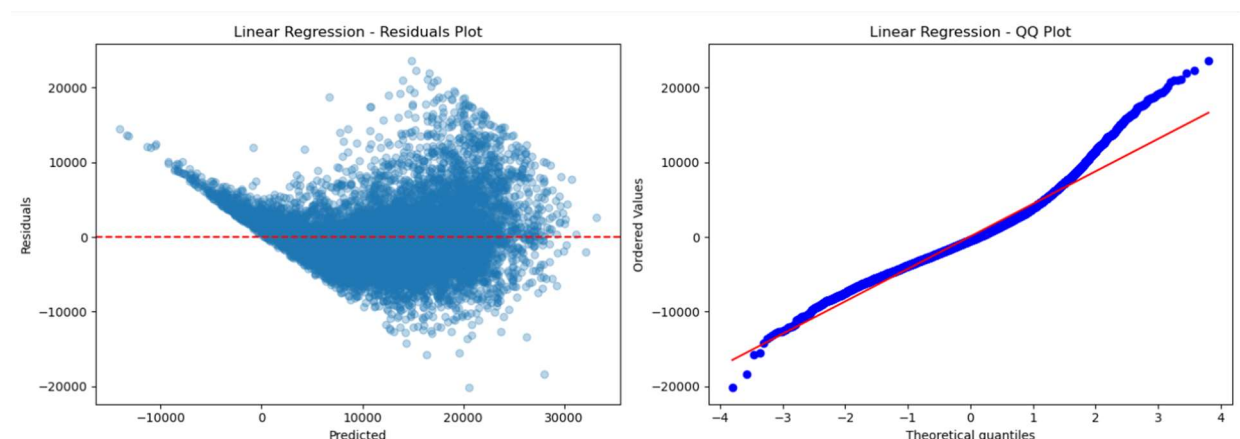
Metrics

RMSE	MAE	R2
4446.57	3303.02	0.706

Plots

Residuals Plot: residuals are scattered around 0 but with high spread, indicating the model struggles with non-linear relationships in the data.

QQ Plot: residuals deviate significantly from the normal distribution line, especially in the tails — confirming the model is not capturing complex patterns.



Model 2: Decision Tree

Decision Tree Regressor - a non-linear model that splits data based on feature thresholds. More flexible than linear regression but prone to overfitting.

Grid Search Parameters

Parameter	Values Tested
max_depth	[5, 10, 20]
min_samples_split	[2, 5]

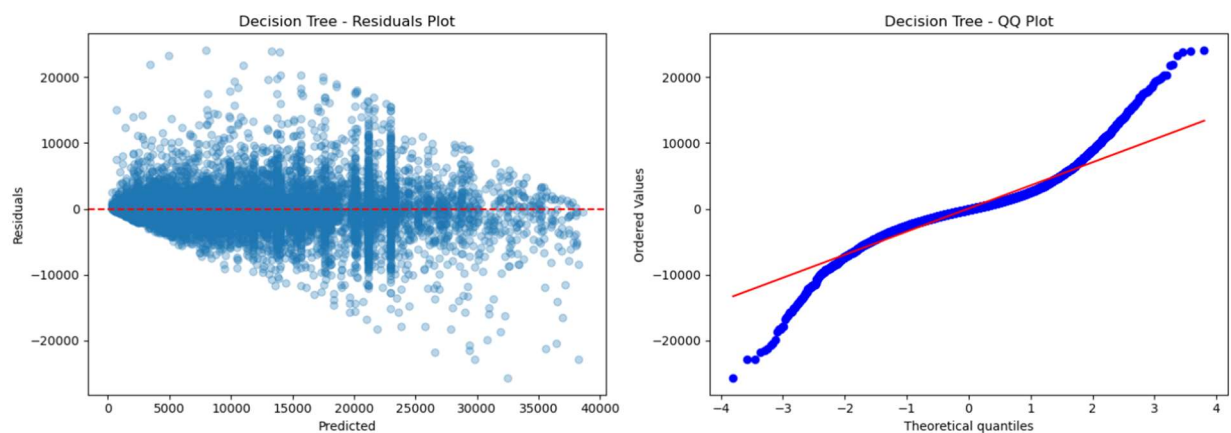
Metrics

RMSE	MAE	R2
3669.11	2427.88	0.800

Plots

Residuals Plot: residuals are more concentrated around 0 compared to Linear Regression, showing better fit. Some systematic patterns remain.

QQ Plot: closer to normal distribution than Linear Regression, but tails still deviate — signs of overfitting on some price ranges.



Model 3: Random Forest

Random Forest Regressor - an ensemble of decision trees trained on random subsets of data and features. More robust and less prone to overfitting than a single tree.

Grid Search Parameters

Parameter	Values Tested
n_estimators	[50, 100]
max_depth	[10, 20]

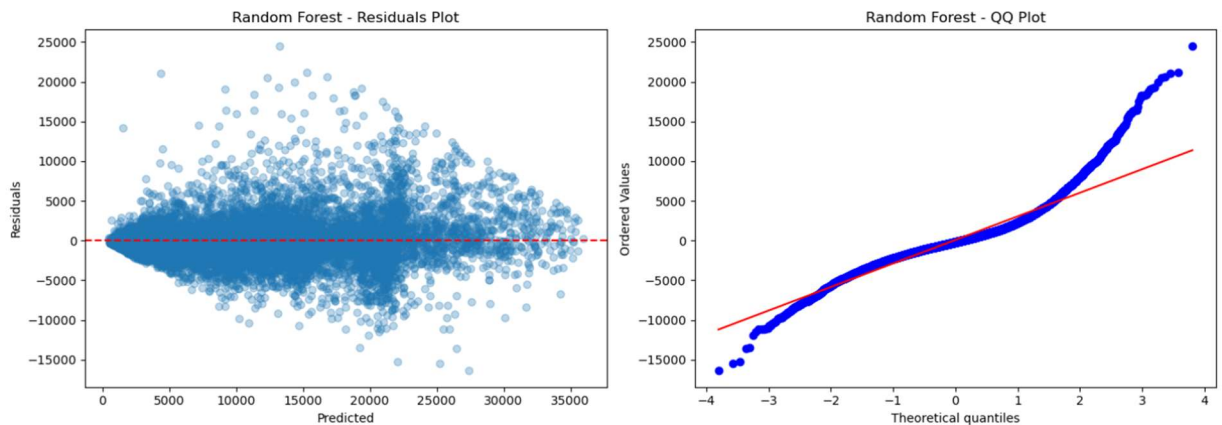
Metrics

RMSE	MAE	R2
3109.13	2097.77	0.857

Plots

Residuals Plot: residuals are well centered around 0 with the smallest spread among all models. No strong systematic patterns visible.

QQ Plot: residuals follow the normal distribution most closely of all four models, confirming good model fit.



Model 4: Gradient Boosting

Gradient Boosting Regressor - an ensemble method that builds trees sequentially, each correcting the errors of the previous one. Strong performance but slower to train.

Grid Search Parameters

Parameter	Values Tested
n_estimators	[50, 100]
max_depth	[3, 5]
learning_rate	[0.05, 0.1]

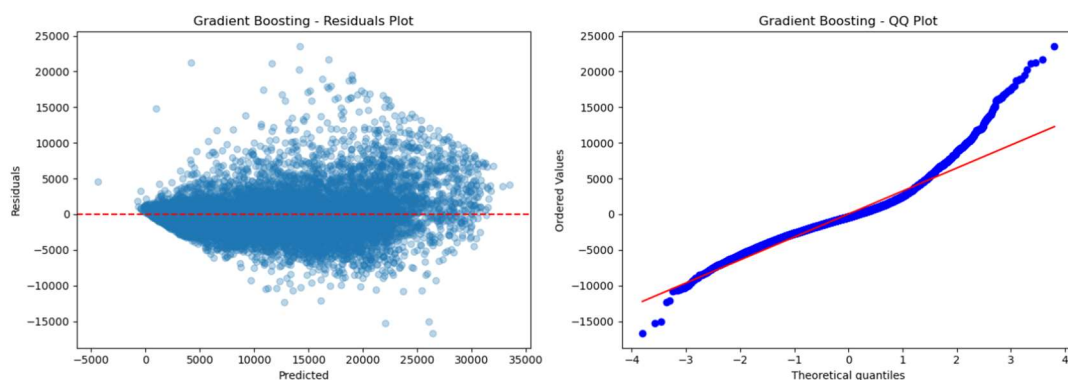
Metrics

RMSE	MAE	R2
3348.66	2356.28	0.833

Plots

Residuals Plot: residuals are well centered, similar to Random Forest. Slight underprediction visible for very high-priced vehicles.

QQ Plot: distribution is close to normal but with slightly heavier tails than Random Forest.



Best Model

Selected Model: Random Forest Regressor

Main metric used for selection: **R2 (coefficient of determination)**

R2 was chosen as the primary metric because it directly measures how much of the variance in selling price is explained by the model, making it easy to interpret and compare across models.

Model	RMSE	MAE	R2
Linear Regression	4446.57	3303.02	0.706
Decision Tree	3669.11	2427.88	0.800
Random Forest	3109.13	2097.77	0.857
Gradient Boosting	3348.66	2356.28	0.833

Reasons for selecting Random Forest:

- Highest R2 = 0.857 - explains 85.7% of the variance in vehicle selling price
- Lowest RMSE = 3109 - predictions are closest to actual prices on average
- Lowest MAE = 2097 - on average, predictions are off by only ~\$2097
- More robust than a single Decision Tree due to ensemble averaging
- Outperforms Gradient Boosting on all three metrics while being faster to train
- Handles non-linear relationships and feature interactions well